Guidance and Control of Unmanned Autonomous Vehicles

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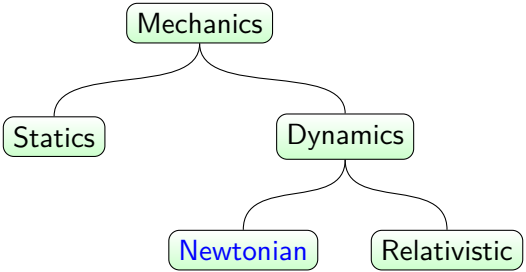
Mechanics: Study of the effect that physical forces have on objects.

Statics: Analysis of loads and their effect on a system when its acceleration is zero or it is static equilibrium with the environment (relative motion is zero).



- **Dynamics**: Study of the effect that forces have on the motion of objects.
  - ▷ **Newtonian dynamics**: Study of the motion of objects that travel with speeds significantly less than the speed of light, $(V \ll c)$
  - ▷ **Relativistic dynamics**: Study of the motion of objects that travel with speeds at or near the speed of light, $(V \approx c)$

Unmanned Autonomous Vehicles

- **Kinematics**: Study of the geometry of motion without regard to the forces that cause the motion.
- **Objectives of kinematics**
  - A set of reference frames in which to observe the motion of a system
  - A set of coordinate systems fixed in the chosen reference frames
  - Angular velocity and angular acceleration of each reference frame (and/or rigid body) resolved in the chosen coordinate systems
  - Position, velocity, and acceleration of each particle in the system
- Any physical motion cannot occur without some kind of external inputs.
- **Kinetics of the particle**:
  - To describe quantitatively the forces that act on a particle
  - To determine the motion that results from the application of these forces using postulated laws of physics
  - To analyze the motion

Unmanned Autonomous Vehicles: Reference frame

- Are reference frame and coordinate system the same? Not really

Reference frame

A perspective from which observations are made regarding the motion of a system.

- $\mathcal{C}$: A collection of at least three noncolinear points that move in three-dimensional Euclidean space $\mathbb{R}^3$.

Rigidly connected or rigidly attached

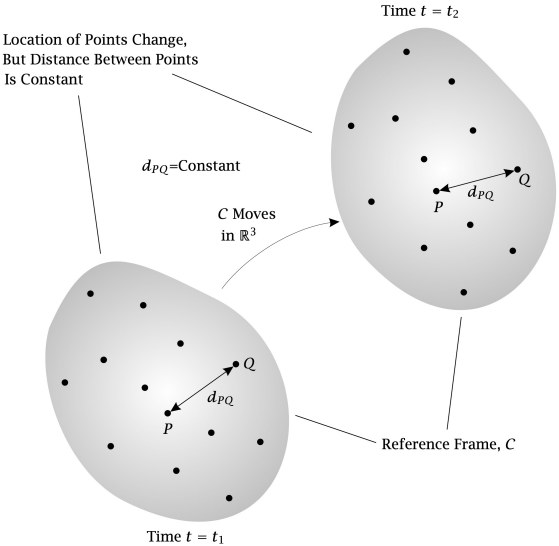
Points P and Q in $\mathcal{C}$ are said to be rigidly connected or rigidly attached if the distance between P and Q, denoted d_{PQ}, is constant regardless of how the points P and Q move in $\mathbb{R}^3$.

Unmanned Autonomous Vehicles: Reference frame

Reference Frame

A collection $\mathcal{C}$ is then said to be a reference frame if the distance between every pair of points in $\mathcal{C}$ is rigidly connected.

Tenenbaum 2004



Unmanned Autonomous Vehicles: Reference frame

Reference frame

A reference frame is a collection of at least three noncolinear points in $\mathbb{R}^3$ such that the distance between any two points in the collection does not change with time.

Tenenbaum 2004

- **Example of reference frames**:
  - Any three-dimensional rigid body (e.g., a cube, a sphere, or a cylinder)
  - Any planar rigid object (e.g., a square, a circle, or a triangle)
- What about an isolated point in $\mathbb{R}^3$? No
- What about a line in $\mathbb{R}^3$? No, why? Points on a line are colinear.

Unmanned Autonomous Vehicles: Invariance of Space and Time

- **Independent variable**: A quantity that parameterizes the motion, and whose value determines the values of all other quantities of interest.
- **Convenient choice**: One that is independent of the reference frame
  - ▷ Time in the Newtonian mechanics
- **Assumption of Newtonian mechanics**: Space and time are invariant with respect to changes in reference frames.
- Observations made of space are the same in all reference frames.
- Observations of an arbitrary vector $\boldsymbol{b}$ as well as time are the same, regardless of the reference frame.
- For any two reference frames $\mathcal{A}$ and $\mathcal{B}$,

$$\boxed{\boldsymbol{b}^{\mathcal{A}} = \boldsymbol{b}^{\mathcal{B}} = \boldsymbol{b}; \quad t^{\mathcal{A}} = t^{\mathcal{B}} = t}$$

Unmanned Autonomous Vehicles: Nature of Reference frames

- **Classification of reference frames**: Inertial or noninertial.
- **Inertial or Newtonian reference frame**: One whose points are either absolutely fixed in space or at most translate relative to an absolutely fixed set of points with the same constant velocity.
- **Noninertial or non-Newtonian reference frame**: One whose points accelerate with time.
- **Axiom of Newtonian mechanics**:
  - The inertial reference frames exist, and the laws of mechanics are valid **only** in an inertial reference frame.
- Any possible difficulties encountered with the reference frame?
- Definition of reference frame does not directly provide any means of measuring observations that may be made of the motion by observer fixed in reference frame.
- **Coordinate system**: To quantify or realize the motion of the system.

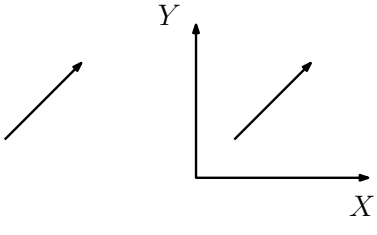
Unmanned Autonomous Vehicles: Coordinate system

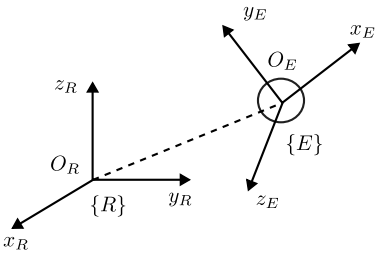
- **Coordinate system**
  - ▷ What are the elements of a coordinate system?
  - ▷ A point (the origin) fixed in reference frame $\mathcal{A}$, from which all distances are measured.
  - ▷ A basis, which is a set of three linearly independent directions that are fixed in reference frame $\mathcal{A}$.
  - ▷ A basis provides a way to resolve the vectors in $\mathbb{R}^3$.
- A coordinate system must be fixed in a unique reference frame.
- A coordinate system cannot be fixed simultaneously in two distinct reference frames.
- How to choose the basis of coordinate frame?
  - ▷ Any set of three linearly independent vectors
  - ▷ Is it convenient? May not be always convenient
  - ▷ A set of mutually orthogonal unit vectors for better convenience

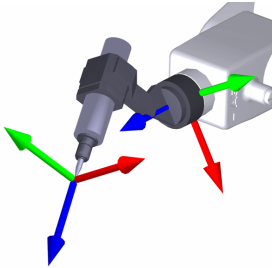
Unmanned Autonomous Vehicles: Coordinate Frame

- What are the reasons for requirements of different coordinate frames?
- Newton's equations of motion are derived relative to a fixed, inertial reference frame. However, the motion is most easily described in a body-fixed frame.
- Aerodynamic forces and torques act on the aircraft body and are most easily described in a body-fixed reference frame.
- On-board sensors like accelerometers and rate gyros measure information w.r.t. the body frame.
- GPS measures position, ground speed, and course angle w.r.t. the inertial frame.
- Most mission requirements, like loiter points and flight trajectories, as well as map information are specified in the inertial frame.
- **Transformation of one frame to another**: Translation and rotation

Coordinate Transformation







- **Position of rigid body**: Position vector $O_R O_E$ of origin
- **Orientation of rigid body**: 3×3 rotation matrix
- For simplification, we assume $O_E = O_R$.

Coordinate Transformation

- Rotation matrix approach utilizes 9 parameters, which obey the orthogonality and unit length constraints, to describe the orientation of the rigid body.
- A rigid body possesses 3 rotational DOF, 3 independent parameters are sufficient to characterize completely and unambiguously its orientation.
- Three-parameter representations are popular in engineering because they minimize the dimensionality of the rigid-body control problem.
- Transformation of coordinate axes is an important necessity in resolving angular positions and rates from one coordinate system to another.
- **Transformation matrix**: Mapping of the components of a vector, resolved in one frame, into the same resolved into the another frame.

  - ▷ Direction cosine matrix (DCM)
  - ▷ Euler Angles
  - ▷ Quaternions
  - ▷ Rotation vector

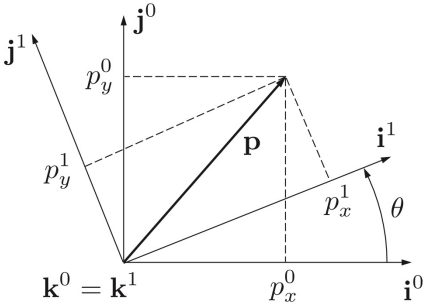
Unmanned Autonomous Vehicles: Coordinate Frame

- Vector $\boldsymbol{p}$ in $\mathcal{F}^0$ frame

$$\boxed{\boldsymbol{p} = p_x^0 \boldsymbol{i}^0 + p_y^0 \boldsymbol{j}^0 + p_z^0 \boldsymbol{k}^0}$$

- Vector $\boldsymbol{p}$ in $\mathcal{F}^1$ frame

$$\boxed{\boldsymbol{p} = p_x^1 \boldsymbol{i}^1 + p_y^1 \boldsymbol{j}^1 + p_z^1 \boldsymbol{k}^1}$$



- Since the vector is same, we have

$$\boxed{p_x^1 \boldsymbol{i}^1 + p_y^1 \boldsymbol{j}^1 + p_z^1 \boldsymbol{k}^1 = p_x^0 \boldsymbol{i}^0 + p_y^0 \boldsymbol{j}^0 + p_z^0 \boldsymbol{k}^0}$$

- If we denote vector representations in $\mathcal{F}^0$ and $\mathcal{F}^1$ as $\boldsymbol{p}^0$ and $\boldsymbol{p}^1$ then $\boxed{\boldsymbol{p}^1 = \mathcal{R}_0^1 \boldsymbol{p}^0}$

Unmanned Autonomous Vehicles: Coordinate Frame

- Taking dot products on both sides with unit vectors $\boldsymbol{i}^1$, $\boldsymbol{j}^1$, and $\boldsymbol{k}^1$,

$$\begin{pmatrix} p_x^1 \\ p_y^1 \\ p_z^1 \end{pmatrix} = \begin{pmatrix} \boldsymbol{i}^1 \cdot \boldsymbol{i}^0 & \boldsymbol{i}^1 \cdot \boldsymbol{j}^0 & \boldsymbol{i}^1 \cdot \boldsymbol{k}^0 \\ \boldsymbol{j}^1 \cdot \boldsymbol{i}^0 & \boldsymbol{j}^1 \cdot \boldsymbol{j}^0 & \boldsymbol{j}^1 \cdot \boldsymbol{k}^0 \\ \boldsymbol{k}^1 \cdot \boldsymbol{i}^0 & \boldsymbol{k}^1 \cdot \boldsymbol{j}^0 & \boldsymbol{k}^1 \cdot \boldsymbol{k}^0 \end{pmatrix} \begin{pmatrix} p_x^0 \\ p_y^0 \\ p_z^0 \end{pmatrix}$$

- General transformation matrix (also called DCM)

$$\mathcal{R}_0^1 = \begin{pmatrix} \boldsymbol{i}^1 \cdot \boldsymbol{i}^0 & \boldsymbol{i}^1 \cdot \boldsymbol{j}^0 & \boldsymbol{i}^1 \cdot \boldsymbol{k}^0 \\ \boldsymbol{j}^1 \cdot \boldsymbol{i}^0 & \boldsymbol{j}^1 \cdot \boldsymbol{j}^0 & \boldsymbol{j}^1 \cdot \boldsymbol{k}^0 \\ \boldsymbol{k}^1 \cdot \boldsymbol{i}^0 & \boldsymbol{k}^1 \cdot \boldsymbol{j}^0 & \boldsymbol{k}^1 \cdot \boldsymbol{k}^0 \end{pmatrix}$$

- Interpretation of DCM in terms of the unit vectors of both frames
- **Unit vector of new frame in terms of old unit vectors, arranged in a row**

Unmanned Autonomous Vehicles: Coordinate Frame

- Transformation matrix for rotation about the $z-$axis

$$\mathcal{R}_0^1(\psi) \triangleq \begin{pmatrix} \cos\psi & \sin\psi & 0 \\ -\sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

- Transformation matrix for rotation about $y-$axis and $x-$axis, respectively,

$$\mathcal{R}_0^1(\theta) \triangleq \begin{pmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{pmatrix}, \quad \mathcal{R}_0^1(\phi) \triangleq \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & \sin\phi \\ 0 & -\sin\phi & \cos\phi \end{pmatrix}$$

- Transformation matrix is an orthogonal rotation matrix.

$$\boxed{\left(\mathcal{R}_a^b\right)^{-1} = \left(\mathcal{R}_a^b\right)^\top = \mathcal{R}_b^a; \quad \mathcal{R}_b^c\mathcal{R}_a^b = \mathcal{R}_a^c; \quad \det\left(\mathcal{R}_a^b\right) = 1}$$

Coordinate Transformation

Consider two coordinate frames with their unit vectors as $(\boldsymbol{i}, \boldsymbol{j}, \boldsymbol{k})$, and $(\boldsymbol{i}', \boldsymbol{j}', \boldsymbol{k}')$, respectively. If $\boldsymbol{i}' = \boldsymbol{j}$, $\boldsymbol{j}' = -\boldsymbol{i}$, and $\boldsymbol{k}' = \boldsymbol{k}$ then what would be the direction cosine matrix (DCM)?

$$\mathcal{C} = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Consider two coordinate frames with their unit vectors as $(\boldsymbol{i}, \boldsymbol{j}, \boldsymbol{k})$, and $(\boldsymbol{i}', \boldsymbol{j}', \boldsymbol{k}')$, respectively. If $\boldsymbol{i}' = \boldsymbol{i}$, $\boldsymbol{j}' = -\boldsymbol{k}$, and $\boldsymbol{k}' = \boldsymbol{j}$ then what would be the direction cosine matrix (DCM)?

$$\mathcal{C} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$$

Coordinate Transformation

Consider two coordinate frames with their unit vectors as $(\boldsymbol{i}, \boldsymbol{j}, \boldsymbol{k})$, and $(\boldsymbol{i}', \boldsymbol{j}', \boldsymbol{k}')$, respectively. If the old coordinate frame is rotated with angle θ in anti-clockwise direction w.r.t. z-axis to get a new frame, then what would be the direction cosine matrix (DCM)?

- Unit vectors of new frame

$$\boxed{\boldsymbol{i}' = \cos\theta \boldsymbol{i} + \sin\theta \boldsymbol{j}, \quad \boldsymbol{j}' = -\sin\theta \boldsymbol{i} + \cos\theta \boldsymbol{j}, \quad \boldsymbol{k}' = \boldsymbol{k}}$$

- DCM matrix

$$\begin{pmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Coordinate Transformation

Find out the missing coefficients of DCM given below

$$T = \begin{bmatrix} 0.8999 & -0.4323 & 0.0578 \\ c_{21} & 0.8665 & -0.2496 \\ c_{31} & c_{32} & 0.9666 \end{bmatrix}$$

- One can use the orthogonal properties of DCM.

$$0.9666c_{32} - 0.8665 \times 0.2496 - 0.4323 \times 0.0578 = 0$$

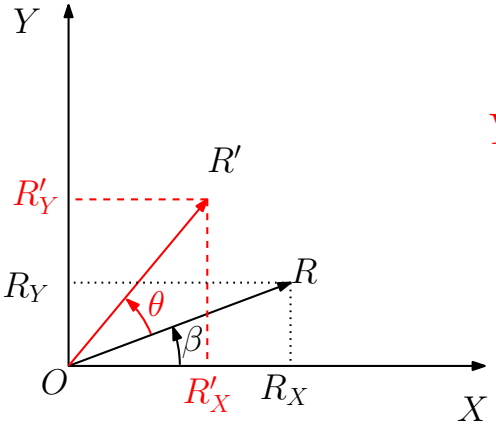
$$c_{31}c_{32} + 0.8665c_{21} - 0.8999 \times 0.4323 = 0$$

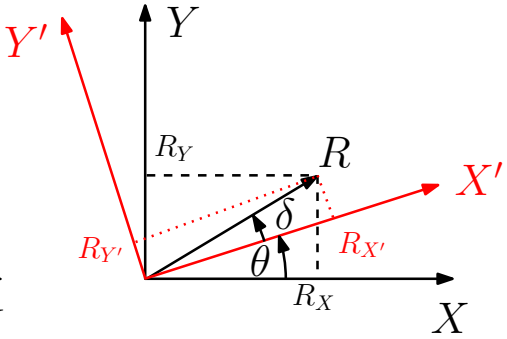
$$0.9666c_{31} - 0.2496c_{21} + 0.0578 \times 0.8999 = 0$$

- On solving, $c_{21} = 0.4323,\ c_{31} = 0.0578,\ c_{32} = 0.2496,$ **Check for correctness**

Coordinate Transformation

How to derive the rotation matrix for rotation in the planar case?





Coordinate Transformation

- The position of a point R in XY coordinate frame is given by

$$\begin{bmatrix} R_X \\ R_Y \end{bmatrix} = \begin{bmatrix} R\cos\beta \\ R\sin\beta \end{bmatrix}$$

- Let us assume $\gamma = \theta + \beta$.
- Position of a point R' in XY coordinate frame is given by

$$\begin{bmatrix} R'_X \\ R'_Y \end{bmatrix} = \begin{bmatrix} R\cos\gamma \\ R\sin\gamma \end{bmatrix} = \begin{bmatrix} R\cos\theta\cos\beta - R\sin\theta\sin\beta \\ R\sin\theta\cos\beta + R\cos\theta\sin\beta \end{bmatrix}$$

$$= \underbrace{\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}}_{\text{Rotation matrix}} \begin{bmatrix} R_X \\ R_Y \end{bmatrix}$$

Coordinate Transformation

- Let us assume $\alpha = \theta + \delta$.
- Position of a point R in XY frame: $\begin{bmatrix} R_X \\ R_Y \end{bmatrix} = \begin{bmatrix} R\cos\alpha \\ R\sin\alpha \end{bmatrix}$
- Position of a point R in $X'Y'$ frame: $\begin{bmatrix} R_{X'} \\ R_{Y'} \end{bmatrix} = \begin{bmatrix} R\cos\delta \\ R\sin\delta \end{bmatrix}$
- As $\boxed{\delta = \alpha - \theta}$, one can also write

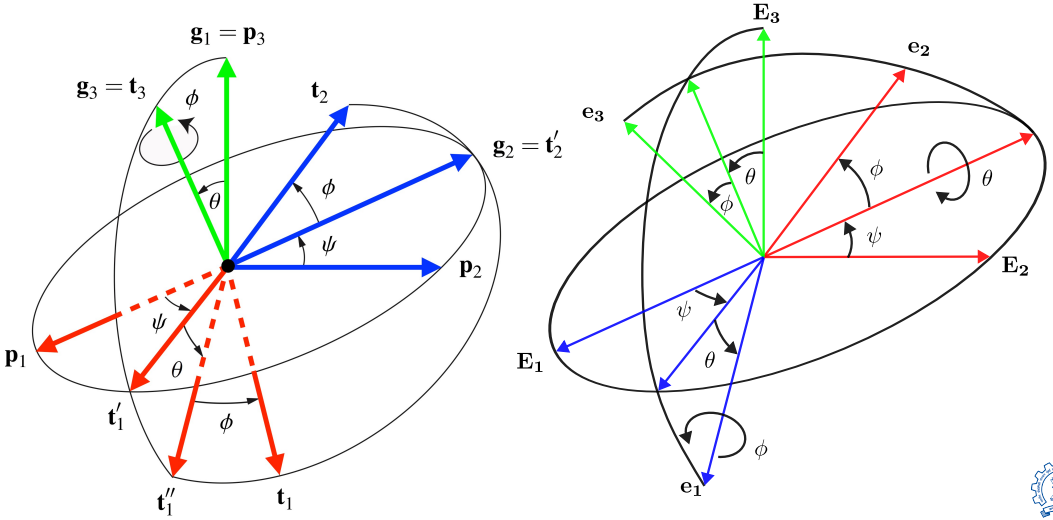
$$\begin{bmatrix} R_{X'} \\ R_{Y'} \end{bmatrix} = \begin{bmatrix} R\cos(\alpha-\theta) \\ R\sin(\alpha-\theta) \end{bmatrix} = \begin{bmatrix} R\cos\alpha\cos\theta + R\sin\alpha\sin\theta \\ R\sin\alpha\cos\theta - R\cos\alpha\sin\theta \end{bmatrix}$$

$$= \begin{bmatrix} R_X\cos\theta + R_Y\sin\theta \\ R_Y\cos\theta - R_X\sin\theta \end{bmatrix} = \underbrace{\begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}}_{\text{Rotation matrix}} \begin{bmatrix} R_X \\ R_Y \end{bmatrix}$$

Coordinate Transformation

- **Euler angles**
  - ▷ Method to specify angular orientation of one coordinate frame w.r.t. another frame
  - ▷ A series of three ordered right-handed rotations
  - ▷ Correspond to the conventional roll-pitch-yaw angles
- Euler angles not uniquely defined since there is an infinite set of choices
- No standardized definitions of the Euler angles
- For a particular choice of Euler angles, the rotation order selected and/or defined should be consistent.
- Interchange in order of rotation ▷ different Euler angle representation
- Rotations are made about the Z, Y, X axes through an angle ψ, θ, ϕ angles.
- These rotations are made in the positive (anticlockwise sense) when **looking down the axis of rotation toward the origin**

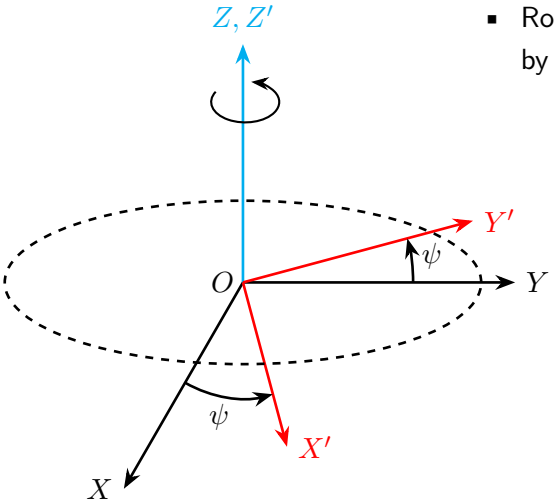
Coordinate Transformation, $(ZY'Z'')$ and $(ZY'X'')$



Coordinate Transformation

- **Euler angles**: Three ordered elemental rotations
- **Extrinsic rotations**: Rotations about the axes xyz of the original coordinate system, which is assumed to remain motionless.
- **Intrinsic rotations**: Rotations about the axes of rotating coordinate system XYZ, which changes its orientation after each elemental rotation.
- Another classification
  - ▷ Proper Euler angles
  - ▷ Tait-Bryan angles
- **Proper Euler angles**: $zxz, zyz, xyx, xzx, yzy, yxy$
- **Tait-Bryan angles**: $zyx, zxy, xyz, xzy, yzx, yxz$
- What is the major difference between Proper Euler and Tait-Bryan angles?
- Tait-Bryan angles: Rotations about three distinct axes
- Proper Euler angles: Same axis for both the first and third elemental rotations

Coordinate Transformation



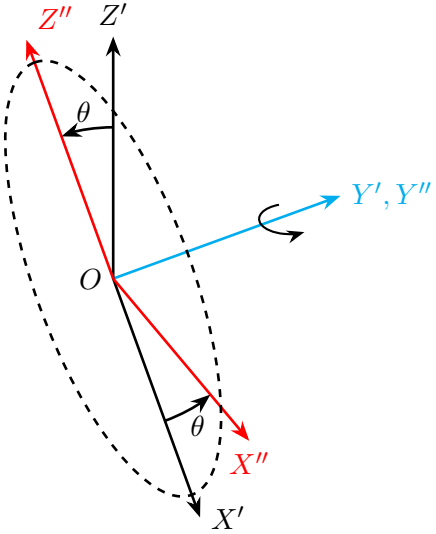
- Rotation about Z axis in <u>anticlockwise direction</u> by an angle ψ

$$\begin{bmatrix} X' \\ Y' \\ Z' \end{bmatrix} = \begin{bmatrix} \cos\psi & \sin\psi & 0 \\ -\sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}$$

$$= \mathcal{A} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}$$

$$\mathcal{A} = \begin{bmatrix} \cos\psi & \sin\psi & 0 \\ -\sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Coordinate Transformation



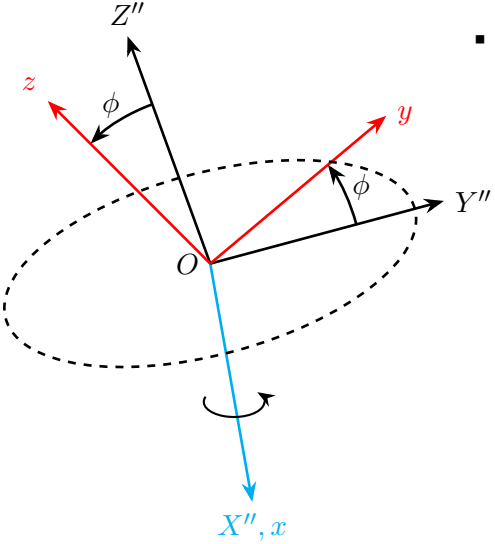
- Rotation about Y axis in <u>anticlockwise direction</u> by an angle θ

$$\begin{bmatrix} X'' \\ Y'' \\ Z'' \end{bmatrix} = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix} \begin{bmatrix} X' \\ Y' \\ Z' \end{bmatrix}$$

$$= \mathcal{B} \begin{bmatrix} X' \\ Y' \\ Z' \end{bmatrix}$$

$$\mathcal{B} = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix}$$

Coordinate Transformation



- Rotation about X axis in <u>anticlockwise direction</u> by an angle ϕ

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & \sin\phi \\ 0 & -\sin\phi & \cos\phi \end{bmatrix} \begin{bmatrix} X'' \\ Y'' \\ Z'' \end{bmatrix}$$

$$= \mathcal{D} \begin{bmatrix} X'' \\ Y'' \\ Z'' \end{bmatrix}$$

$$\mathcal{D} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & \sin\phi \\ 0 & -\sin\phi & \cos\phi \end{bmatrix}$$

Coordinate Transformation

- If the consecutive rotations are performed in the order ψ, θ, ϕ, i.e., (yaw, pitch, and roll) on reference frame XYZ, then we obtain another reference frame xyz.
- Rotation matrix for representing these three rotations

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \mathcal{D} \begin{bmatrix} X'' \\ Y'' \\ Z'' \end{bmatrix} = \mathcal{DB} \begin{bmatrix} X' \\ Y' \\ Z' \end{bmatrix} = \mathcal{DBA} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}$$

- Equivalently,

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \underbrace{\mathcal{DBA}}_{\mathcal{C}} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}$$

Coordinate Transformation

- Equivalent rotation matrix $\mathcal{C} = \mathcal{DBA}$ can be written as

$$\mathcal{C} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & \sin\phi \\ 0 & -\sin\phi & \cos\phi \end{bmatrix} \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix} \begin{bmatrix} \cos\psi & \sin\psi & 0 \\ -\sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \cos\theta\cos\psi & \cos\theta\sin\psi & -\sin\theta \\ \cos\psi\sin\theta\sin\phi - \sin\psi\cos\phi & \sin\psi\sin\theta\sin\phi + \cos\psi\cos\phi & \cos\theta\sin\phi \\ \cos\psi\sin\theta\cos\phi + \sin\psi\sin\phi & \sin\psi\sin\theta\cos\phi - \cos\psi\sin\phi & \cos\theta\cos\phi \end{bmatrix}$$

- Rotation matrix: **Euler angle transformation matrix**
- **Range of Euler angles**: $\boxed{-\pi \leq \psi \leq \pi, \quad -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}, \quad -\pi \leq \phi \leq \pi}$
- Is there any issue with range $|\theta| > \frac{\pi}{2}$?

Coordinate Transformation

- Equivalent rotation matrix

$$\mathcal{C} = \begin{bmatrix} \cos\theta\cos\psi & \cos\theta\sin\psi & -\sin\theta \\ \cos\psi\sin\theta\sin\phi - \sin\psi\cos\phi & \sin\psi\sin\theta\sin\phi + \cos\psi\cos\phi & \cos\theta\sin\phi \\ \cos\psi\sin\theta\cos\phi + \sin\psi\sin\phi & \sin\psi\sin\theta\cos\phi - \cos\psi\sin\phi & \cos\theta\cos\phi \end{bmatrix}$$

$$= \begin{bmatrix} \mathcal{C}_{11} & \mathcal{C}_{12} & \mathcal{C}_{13} \\ \mathcal{C}_{21} & \mathcal{C}_{22} & \mathcal{C}_{23} \\ \mathcal{C}_{31} & \mathcal{C}_{32} & \mathcal{C}_{33} \end{bmatrix}$$

- How to obtain Euler angles from a given DCM matrix?

$$\theta = \sin^{-1}(-\mathcal{C}_{13}), \quad \phi = \sin^{-1}\left(\frac{\mathcal{C}_{23}}{\sqrt{1-\mathcal{C}_{13}^2}}\right), \quad \psi = \sin^{-1}\left(\frac{\mathcal{C}_{12}}{\sqrt{1-\mathcal{C}_{13}^2}}\right)$$

- Are there some issues with these expressions?

Text/References

References

- George M. Siouris, Aerospace Avionics Systems: A Modern Synthesis, Academic Press, Inc. 1993.
- Bandhu N. Pamadi, Performance, Stability, and Control of Airplanes, AIAA Education Series, 1998.

Thank you for your attention !!!